Edge Computing vs Cloud Computing

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Technical Presentation

Introduction

What is Edge Computing?

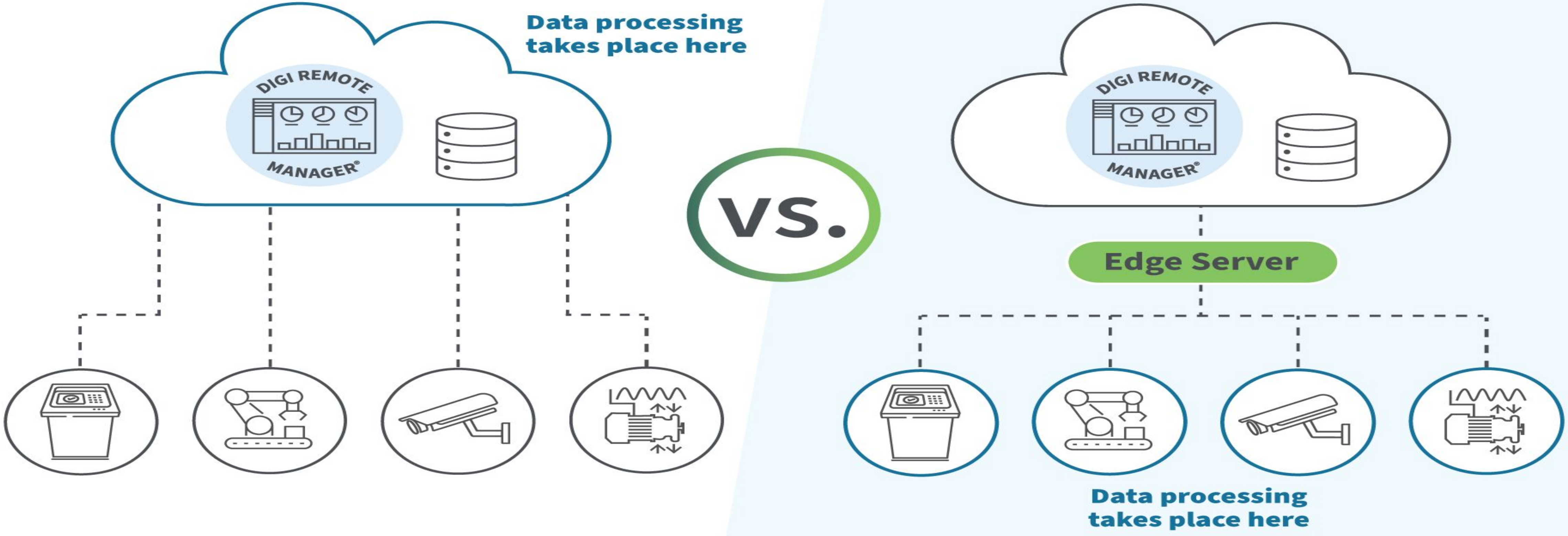
- Edge computing processes data near the source where it is generated.
- Reduces latency and bandwidth usage.
- Suitable for real-time applications.

What is Cloud Computing?

- Cloud computing processes and stores data in centralized remote data centers.
- Provides scalable computing resources over the internet.
- Supports large-scale data storage and analytics.

TRADITIONAL
Cloud Computing

MODERN
Edge Computing



Core Technical Explanation

Edge Computing

- Data is processed locally on edge devices or nearby edge servers.
- Only important data is sent to the cloud.
- Enables faster response time.

Cloud Computing

- Data is transmitted to centralized cloud servers.
- Cloud performs processing, storage, AI, and analytics.
- Offers virtually unlimited computing resources.

BENIFITS

Edge Computing

- Ultra-low latency
- Faster decision making
- Reduced bandwidth usage
- Improved privacy
- Works during limited internet connectivity

Cloud Computing

- High scalability
- Cost-efficient infrastructure
- Powerful computing resources
- Global accessibility
- Easy backup and disaster recovery

Real-World Applications

Netflix – Cloud Computing

Stores and streams millions of movies and TV shows worldwide.

Uses cloud infrastructure to scale based on user demand.

Autonomous Vehicles (Tesla, Waymo) – Edge Computing

Self-driving cars process sensor and camera data locally.

Decisions like braking, steering, and obstacle detection require milliseconds.

Smart Manufacturing (Siemens, Bosch) – Edge + Cloud

Factory machines monitor temperature, pressure, and vibration.

Edge devices detect equipment failures instantly.

Cloud stores production data for long-term analysis.

Advantages

- Reduced traffic congestion
- Faster emergency response
- Better fuel efficiency
- Improved road safety

Conclusion

- Edge computing provides fast local processing with minimal latency.
- Cloud computing offers powerful centralized storage and computing.
- Many modern systems combine Edge and Cloud for better performance.
- The right computing paradigm depends on application requirements.

Engineer to Excel

Thank You!!!